Yuyan Chen

Montreal, QC
yuyan-c.github.io | yuyan-chen | Yuyan-C | yuyan.chen2@mail.mcgill.ca

Education

McGill University, Mila - Quebec AI Institute 2023 – Present
Doctor of Philosophy in Computer Science, Advisor: David Rolnick *Montreal, Canada*

McGill University 2019 – 2023
Bachelor of Science, Honours Math and Computer Science, Minor in Environment *Montreal, Canada*

Research Experience

Automated Species Description Jan. 2026 – Present
Mila and Agriculture and Agri-Food Canada

- Building vision language models to generate detailed taxonomic description of new species

Koa Restoration Jan. 2025 – Present
Ohio State University and National Ecological Observatory Network (NEON)

- Deployed camera traps and microphones and conducted drone survey to study the effectiveness of koa restoration in Pu'u Maka'ala Natural Area Reserve, Hawai'i
- Developing machine learning based pipelines to automate data processing

Automated Monitoring of Insects May 2023 – Present
Mila - Quebec AI Institute and Montreal Insectarium

- Using computer vision to discover new insect species in camera-trap images
- Developing open-set recognition and category discovery algorithms

Machine Learning Storage Benchmarking Aug. 2022 – Nov. 2022
McGill University

- Aimed at building an accelerator-agnostic benchmark for storage in ML workloads
- Responsible for data augmentation and generation for different workloads

Literature Analysis of Pest Invasions Sept. 2021 – Dec. 2021
McGill University

- Automation of literature searching, screening, downloading, and keyword highlighting with Python and R
- Used latent Dirichlet allocation (LDA) for topic modeling of papers, reviews, and government reports on invasive forest pest management

Publications

* denotes equal contribution.

Open-Insect: Benchmarking Open-Set Recognition of Novel Species in Biodiversity Monitoring
Neural Information Processing Systems (NeurIPS 2025) Datasets and Benchmarks Track (Spotlight).
Chen, Y., Lang, N., Schmidt, B. C., Jain, A., Basset, Y., Beery, S., Larrivée, M., & Rolnick, D.

Scaling up multi-sensor monitoring of forest restoration with computer vision and deep learning: A case study from Hawai'i

International Conference on Computer Vision (ICCV 2025) Workshop on Computer Vision for Ecology.

Chen, Y.*, Gabeff, V.*, Jousse, M.*, Zolotarev, F.*, Nepovinnykh, E., Long, M., & Rubenstein, D.

Tracking phenological status and ecological interactions in a Hawaiian cloud forest understory using low-cost camera traps and visual foundation models

2025 Workshop on Camera Traps, AI, and Ecology.

Potlapally, A., Meyers, L., Chen, Y., Long, M., Berger-Wolf, T., Subramoni, H., Megret, R., & Rubenstein, D.

Five organizing themes for invasive forest insect and disease management in Canada and the United States

Forest Ecology and Management, 566, 122046.

Hudgins, E. J., Leung, B., MacQuarrie, C. J. K., McCullough, D. G., Francis, A., Lovett, G. M., Guo, Q., Potter, K. M., Cullingham, C. I., Koch, F. H., Bergman, J. N., Binley, A. D., Robichaud, C., Henry, M., Chen, Y., & Bennett, J. R.

Understanding insect range shifts with out-of-distribution detection

NeurIPS 2023 Workshop on Tackling Climate Change with Machine Learning.

Chen, Y., & Rolnick, D.

Presentations & Talks

2026	**McGill Interfaculty Sustainability Summit** McGill University
2026	**Sustainability Reading Group** Mila - Quebec AI Institute
2025	**Automated Monitoring of Insects Symposium** Aarhus, Denmark
2024	**Joint Annual Meeting of the Entomological Societies of Quebec and Canada** Quebec City, Canada
2024	**ALL IN** Montreal, Canada
2023	**Undergraduate Poster Showcase** McGill University
2022	**Undergraduate Computer Science Research Symposium** McGill University
2022	**Soup and Science** McGill University

Teaching & Mentorship

2024	**Hackathon Mentor** Girls Who Code, McGill University
2024	**Mentor** Bridge Mentorship Porgram, Computer Science, McGill University
2023	**Mentor** Scientista, McGill University
2022 – 2024	**Teaching Assistant** McGill Universit
2020 – 2022	**Undergraduate Mentor** McGill University

Academic Service

2026	**Reviewer** International Conference on Learning Representations (ICLR)
2024	**Reviewer** International Journal of Computer Vision (IJCV)

Leadership & Volunteering

2024 – 2026	**Equity, Diversity, and Inclusion (EDI) Commissioner** Computer Science Graduate Society, McGill University
2025 – Present	**Organizer** Sustainability Reading Group, Mila - Quebec AI Institute
2025	**Volunteer** AI & Climate Talks, Mila - Quebec AI Institute
2024 – Present	**Committee Member** Sustainability Committee, Mila - Quebec AI Institute
2024	**Volunteer** Workshop on Harms and Risks of AI in the Military, Montreal, Canada

Awards

2026	**Excellence Scholarships - Women in AI** Mila - Quebec AI Institute
2024	**Master Research Scholarship** Fonds de Recherche du Québec, Ranked: 5/ 30
2023	**Graduate Excellence Award** McGill University
2023	**First-Class Honours** McGill University
2023	**Edward Rosenthall Memorial Prize in Mathematics** McGill University
2022, 2023	**Science Undergraduate Research Awards ($\times$ 2)** McGill University
2021	**William Macdonald Scholarship in Science** McGill University
2020 - 2022	**Tomlinson Engagement Awardee for Mentoring ($\times$ 5)** McGill University
2020 - 2023	**Dean's Honour List ($\times$ 4)** McGill University